Algorithm 1: MMDAgg $\Delta_\alpha^{\Lambda^w, B_{1:3}}$

Inputs:

- samples $\mathbb{X}_m = (x_i)_{1\leq i\leq m}$ in $\mathbb{R}^d$ and $\mathbb{Y}_n = (y_j)_{1\leq j\leq n}$ in $\mathbb{R}^d$
- choice between permutations (Equation (3)) or wild bootstrap (Equation (6))
- one-dimensional kernels $K_1, \dots, K_d$ satisfying the properties presented in Section 3.1
- level $\alpha \in (0, e^{-1})$
- finite collection of bandwidths Λ in $(0,\infty)^d$
- collection of positive weights $(w_\lambda)_{\lambda\in\Lambda}$ satisfying $\sum_{\lambda\in\Lambda} w_\lambda \leq 1$
- number of simulated test statistics B_1 to estimate the quantiles
- number of simulated test statistics B_2 to estimate the level correction
- number of iterations B_3 for the bisection method

Procedure:

Step 1: compute all simulated test statistics (see Appendix C for a more efficient Step 1)

for $\ell = 1, 2$ **and** $b = 1, \dots, B_\ell$:
  generate $\mu^{(b,\ell)} \sim r$ as in Sections 3.2.1 or 3.2.2 (permutations or Rademacher)
  for $\lambda \in \Lambda$:
    compute $\widehat{M}_{\lambda,\ell}^{b} \coloneqq \widehat{M}_{\lambda}^{\mu^{(b,\ell)}}$ as in Equations (10) or (11)

for $\lambda \in \Lambda$:
  compute $\widehat{M}_{\lambda,1}^{B_1+1} \coloneqq \widehat{\mathrm{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n)$ as in Equations (3) or (6)
  $\left(\widehat{M}_{\lambda,1}^{\bullet 1}, \dots, \widehat{M}_{\lambda,1}^{\bullet B_1+1}\right) = \texttt{sort\_by\_ascending\_order}\left(\widehat{M}_{\lambda,1}^{1}, \dots, \widehat{M}_{\lambda,1}^{B_1+1}\right)$

Step 2: compute $\widehat{u}_\alpha$ using the bisection method

$u_{\min} \coloneqq 0$ **and** $u_{\max} \coloneqq \min_{\lambda\in\Lambda} w_\lambda^{-1}$

repeat B_3 **times**:
  compute $u \coloneqq \frac{u_{\min}+u_{\max}}{2}$
  compute $P_u \coloneqq \frac{1}{B_2}\sum_{b=1}^{B_2} \mathbb{1}\left(\max_{\lambda\in\Lambda}\left(\widehat{M}_{\lambda,2}^{b} - \widehat{M}_{\lambda,1}^{\bullet\lceil (B_1+1)(1-uw_\lambda)\rceil}\right) > 0\right)$
  if $P_u \leq \alpha$ **then** $u_{\min} \coloneqq u$ **else** $u_{\max} \coloneqq u$

$\widehat{u}_\alpha \coloneqq u_{\min}$

Step 3: output test result

if $\widehat{M}_{\lambda,1}^{B_1+1} > \widehat{M}_{\lambda,1}^{\bullet\lceil (B_1+1)(1-\widehat{u}_\alpha w_\lambda)\rceil}$ **for some** $\lambda \in \Lambda$:
  return 1 (reject $\mathcal{H}_0$)
else:
  return 0 (fail to reject $\mathcal{H}_0$)

Time complexity:[6] $\mathcal{O}\big(|\Lambda|(B_1+B_2)(m+n)^2\big)$

Space complexity: $\mathcal{O}\big((m+n)^2 + (B_1+B_2)(m+n)\big)$

6. The time complexity is actually $\mathcal{O}\big(|\Lambda|(B_1+B_2)(m+n)^2 + |\Lambda|B_1\ln(B_1) + |\Lambda|B_2B_3\big)$ which under the reasonable assumption $m+n > \max\left(\sqrt{\ln(B_1)}, \sqrt{B_3}\right)$ gives $\mathcal{O}\big(|\Lambda|(B_1+B_2)(m+n)^2\big)$.

Consider some $u \in \big(0, \min_{\lambda\in\Lambda} w_\lambda^{-1}\big)$. Note that if a single MMD test $\Delta_{uw_\lambda}^{\lambda,B_1}$ has a large associated weight w_λ, then its adjusted level uw_λ is bigger and so the estimated conditional quantile $\widehat{q}_{1-uw_\lambda}^{\lambda,B_1}$ is smaller, which means that we reject this single test more often. Recall that if a single MMD test rejects the null hypothesis, then the aggregated MMDAgg test necessarily rejects the null as well. It follows that a single test $\Delta_{uw_\lambda}^{\lambda,B_1}$ with large weight w_λ is viewed as more important than the other tests in the aggregated procedure. When running an experiment, putting weights on the bandwidths of the single MMD tests can be seen as incorporating prior knowledge about which bandwidths might be better suited to this specific experiment. The choice of prior, or equivalently of weights, is further explored in Section 5.1.

As presented in Section 3.2, the p-value of one of the single MMD tests can be computed as

$$p_{\mathrm{val}}^{\lambda} \coloneqq \frac{1}{B_1+1}\left(1+\sum_{b=1}^{B_1}\mathbb{1}\Big(\widehat{M}_{\lambda,1}^{b} \geq \widehat{M}_{\lambda,1}^{B_1+1}\Big)\right)$$

and, with its adjusted level $\widehat{u}_\alpha^{B_{2:3}} w_\lambda$, it satisfies the property that

$$p_{\mathrm{val}}^{\lambda} \leq \widehat{u}_\alpha^{B_{2:3}} w_\lambda \quad\Longleftrightarrow\quad \widehat{\mathrm{MMD}}_\lambda^2(\mathbb{X}_m,\mathbb{Y}_n) > \widehat{q}_{1-\widehat{u}_\alpha^{B_{2:3}} w_\lambda}^{\lambda,B_1}.$$

Hence, our aggregated MMDAgg test can also be expressed in terms of p-values as

$$\begin{aligned}\Delta_\alpha^{\Lambda^w,B_{1:3}}(\mathbb{X}_m,\mathbb{Y}_n,\mathbb{Z}_{B_1},\mathbb{Z}_{B_2}) &= \mathbb{1}\Big(\widehat{\mathrm{MMD}}_\lambda^2(\mathbb{X}_m,\mathbb{Y}_n) > \widehat{q}_{1-\widehat{u}_\alpha^{B_{2:3}} w_\lambda}^{\lambda,B_1} \text{ for some } \lambda\in\Lambda\Big)\\ &= \mathbb{1}\Big(p_{\mathrm{val}}^{\lambda} \leq \widehat{u}_\alpha^{B_{2:3}} w_\lambda \text{ for some } \lambda\in\Lambda\Big).\end{aligned}$$

We now show that MMDAgg indeed has non-asymptotic level α. We emphasize the non-asymptotic nature of our aggregated test, which allows for the use of MMDAgg even in settings with small fixed sample sizes, where other asymptotic tests (such as the original MMD test of Gretton et al., 2012a) fail to control correctly the probability of type I error.

Proposition 8 (proof in Appendix E.8) *Consider $\alpha \in (0,1)$ and $B_1, B_2, B_3 \in \mathbb{N}\setminus\{0\}$. For a collection Λ of bandwidths in $(0,\infty)^d$ and a collection of positive weights $(w_\lambda)_{\lambda\in\Lambda}$ satisfying $\sum_{\lambda\in\Lambda} w_\lambda \leq 1$, the MMDAgg test $\Delta_\alpha^{\Lambda^w,B_{1:3}}$ has non-asymptotic level α, that is*

$$\mathbb{P}_{p\times p\times r\times r}\big(\Delta_\alpha^{\Lambda^w,B_{1:3}}(\mathbb{X}_m,\mathbb{Y}_n,\mathbb{Z}_{B_1},\mathbb{Z}_{B_2})=1\big) \leq \alpha$$

for all probability density functions p on $\mathbb{R}^d$.

3.6 Uniform separation rate of MMDAgg over Sobolev balls

In this section, we compute the uniform separation rate of our MMDAgg test $\Delta_\alpha^{\Lambda^w,B_{1:3}}$ over the Sobolev ball $\mathcal{S}_d^s(R)$. We then present a collection Λ^w of bandwidths and associated weights for which our aggregated test $\Delta_\alpha^{\Lambda^w,B_{1:3}}$ is almost optimal in the minimax sense.

First, as part of the proof of Theorem 9 in Equation (25), we have shown that the following bound holds

$$\mathbb{P}_{p\times q\times r\times r}\big(\Delta_\alpha^{\Lambda^w,B_{1:3}}(\mathbb{X}_m,\mathbb{Y}_n,\mathbb{Z}_{B_1},\mathbb{Z}_{B_2})=0\big) \leq \frac{\beta}{2} + \min_{\lambda\in\Lambda}\ \mathbb{P}_{p\times q\times r}\Big(\Delta_{\alpha w_\lambda/2}^{\lambda,B_1}\big(\mathbb{Z}_{B_1}\big|\mathbb{X}_m,\mathbb{Y}_n\big)=0\Big).$$

This means that we can control the probability of type II error of our MMDAgg test $\Delta_\alpha^{\Lambda^w,B_{1:3}}$ by controlling the smallest probability of type II error of the single MMD tests $\big(\Delta^{\lambda,B_1}_{\alpha w_\lambda/2}\big)_{\lambda\in\Lambda}$ with adjusted levels. Hence, given a collection Λ of bandwidths with its associated weights $(w_\lambda)_{\lambda\in\Lambda}$, if for some $\lambda\in\Lambda$ the single MMD test $\Delta^{\lambda,B_1}_{\alpha w_\lambda/2}$ has probability of type II error upper bounded by $\beta/2\in(0,0.5)$, then the probability of type II error of our aggregated MMDAgg test $\Delta_\alpha^{\Lambda^w,B_{1:3}}$ is at most β. Intuitively, this means that even if our collection of single MMD tests consists of only one 'good' test (in the sense that it has high power with adjusted level) and many other 'bad' tests (in the sense that they have low power with adjusted levels), MMDAgg would still have high power. This is because when the 'good' MMD test rejects the null hypothesis, MMDAgg also necessarily rejects it. Another point of view on this is that we do not lose any power by testing a wider range of bandwidths as long as the weight of the 'best' test remains the same.

The uniform separation rate of our MMDAgg test $\Delta_\alpha^{\Lambda^w,B_{1:3}}$ over the Sobolev ball $\mathcal{S}_d^s(R)$ is then at most twice the lowest of the uniform separation rates of the single MMD tests $\big(\Delta^{\lambda,B_1}_{\alpha w_\lambda/2}\big)_{\lambda\in\Lambda}$. Combining this result with Theorem 6, we obtain the following upper bound on the uniform separation rate of MMDAgg $\Delta_\alpha^{\Lambda^w,B_{1:3}}$ over the Sobolev ball $\mathcal{S}_d^s(R)$.

Theorem 9 (proof in Appendix E.9) *Consider a collection Λ of bandwidths in $(0,\infty)^d$ such that $\lambda_1\cdots\lambda_d\le 1$ for all $\lambda\in\Lambda$ and a collection of positive weights $(w_\lambda)_{\lambda\in\Lambda}$ such that $\sum_{\lambda\in\Lambda}w_\lambda\le 1$. We assume $\alpha\in(0,e^{-1})$, $\beta\in(0,1)$, $s>0$, $R>0$, $M>0$ and $B_1,B_2,B_3\in\mathbb{N}$ satisfying $B_1\ge\big(\max_{\lambda\in\Lambda}w_\lambda^{-2}\big)\frac{12}{\alpha^2}\big(\log\big(\frac{8}{\beta}\big)+\alpha(1-\alpha)\big)$, $B_2\ge\frac{8}{\alpha^2}\ln\big(\frac{2}{\beta}\big)$ and $B_3\ge\log_2\big(\frac{4}{\alpha}\min_{\lambda\in\Lambda}w_\lambda^{-1}\big)$. The uniform separation rate of the aggregated MMDAgg test $\Delta_\alpha^{\Lambda^w,B_{1:3}}$ over the Sobolev ball $\mathcal{S}_d^s(R)$ can be upper bounded as follows*

$$\rho\big(\Delta_\alpha^{\Lambda^w,B_{1:3}},\mathcal{S}_d^s(R),\beta,M\big)^2\le C_6(M,d,s,R,\beta)\min_{\lambda\in\Lambda}\left(\sum_{i=1}^d\lambda_i^{2s}+\frac{\ln\big(\frac{1}{\alpha}\big)+\ln\Big(\frac{1}{w_\lambda}\Big)}{(m+n)\sqrt{\lambda_1\cdots\lambda_d}}\right)$$

for some positive constant $C_6(M,d,s,R,\beta)$.

We recall from Corollary 7 that the optimal choice of bandwidth $\lambda_i^*=(m+n)^{-2/(4s+d)}$, $i=1,\dots,d$, for the single MMD test $\Delta_\alpha^{\lambda^*,B}$ leads to a uniform separation rate over the Sobolev ball $\mathcal{S}_d^s(R)$ of order $(m+n)^{-2s/(4s+d)}$ which is optimal in the minimax sense. However, this choice depends on the unknown smoothness parameter s and so the test cannot be run in practice with this bandwidth. We now propose a specific choice of collection Λ^w of bandwidths and associated weights, which does not depend on s, and derive the uniform separation rate over the Sobolev ball $\mathcal{S}_d^s(R)$ of our aggregated MMDAgg test $\Delta_\alpha^{\Lambda^w,B_{1:3}}$ using that collection. Intuitively, the main idea is to construct a collection of bandwidths which includes a bandwidth (denoted λ^*) with the property that

$$\frac{1}{a}\left(\frac{m+n}{\ln(\ln(m+n))}\right)^{-2/(4s+d)}\le\lambda_i^*\le\left(\frac{m+n}{\ln(\ln(m+n))}\right)^{-2/(4s+d)}$$

for some $a>1$ and for $i=1,\dots,d$. The extra iterated logarithmic term comes from the additional weight term $\ln\big(\frac{1}{w_\lambda}\big)$ in Theorem 9.